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A Regularization Approach to Joint Blur Identification and Image Restoration

Yu-Li You and M. Kaveh, *Fellow, IEEE*

Abstract— The primary difficulty with blind image restoration, or joint blur identification and image restoration, is insufficient information. This calls for proper incorporation of *a priori* knowledge about the image and the point-spread function (PSF). A well-known space-adaptive regularization method for image restoration is extended to address this problem. This new method effectively utilizes, among others, the piecewise smoothness of both the image and the PSF. It attempts to minimize a cost function consisting of a restoration error measure and two regularization terms (one for the image and the other for the blur) subject to other hard constraints. A scale problem inherent to the cost function is identified, which, if not properly treated, may hinder the minimization/blind restoration process. Alternating minimization is proposed to solve this problem so that algorithmic efficiency as well as simplicity is significantly increased. Two implementations of alternating minimization based on steepest descent and conjugate gradient methods are presented. Good performance is observed with numerically and photographically blurred images, even though no stringent assumptions about the structure of the underlying blur operator is made.

I. INTRODUCTION

A FUNDAMENTAL issue in image restoration is blur removal in the presence of observation noise. Blur may be introduced by relative motion between a camera and the object, by a camera that is out of focus, by atmospheric turbulence, and other factors. Noise may come in the form of thermal noise, measurement errors, or introduced by recording medium, transmission medium, and digitization. Many methods have been reported for restoring the original image under the assumption that the blur operation is exactly known (see, e.g., [1] or [2]). The basic operation involved is simply a deconvolution process that faces the usual difficulties related to the ill conditioning of the blur operation. Many of the proposed methods are, therefore, structured in the context of regularization procedures to make the inversion well behaved. The blur operation is, however, most likely unknown in practice. A more realistic and challenging problem is, therefore, restoration under the condition that the blur operation is unknown and is posed as a function—point spread function (PSF)—to be identified either initially or as an integral part of the restoration procedure.

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The authors are with the Department of Electrical Engineering, University of Minnesota, Minneapolis, MN 55455 USA (email: kaveh@ee.umn.edu).

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Early work on blur identification focused on inherent regular patterns of zeros due to the PSF in the frequency domain [3]–[5]. These are limited to simple blurs, such as motion blur, and are not effective in the presence of reasonable noise levels. More recent work has been based on stochastic linear models for the original image through modeling the distorted image as a 2-D autoregressive moving average (ARMA) process in which the AR parameters are due to the original image and the MA parameters correspond to the PSF [6], [7], [5]. Under some assumptions on the statistics of the driving noise for the process, the parameters can be estimated by maximum-likelihood procedures (ML) using computational approaches such as 2-D recursive estimation [6]–[8], [5], Kalman filtering [9], or expectation maximization (EM) [10]–[12]. Their performance, however, suffers from the restriction of stationarity placed on the image by the AR model and the assumptions that the input and the observation noise are white and Gaussian. The ML function is highly nonlinear, so the algorithm complexity and computational load are significant.

A possible alternative to the above estimators is provided by the generalized cross validation (GCV) method [13], which is motivated by the success of GCV in identifying optimal regularization parameters for regularized image restoration [14]. Also using the ARMA model for the degraded image, GCV determines the parameters that minimize a weighted sum of prediction errors. The prediction is based on space-invariant regularized image restoration. The GCV is reported to often outperform ML in identifying blur and image model parameters [13]. However, its performance also suffers from the assumption of a stationary image, and its algorithmic complexity and computational load are still high.

There is another line of research that started with the iterative loop developed in [15]. Each loop starts with an initial estimate of the original image satisfying a set of constraints (positivity and/or size). This initial estimate and the observed image are used to give an estimate of the PSF through inverse filtering. After imposing constraints (positivity and/or size) on the PSF estimate, it is used along with the observed image to give a new estimate of the original image again using inverse filtering, and constraints are then imposed on it. This completes a loop and the loop is then repeated. A Wiener filter is proposed by [16] to replace the inverse filter used in the loop. In view of the lack of rigorous justification for the convergence of the iterative loop, an alternative is proposed in [17] that minimizes an error measure (energy of restoration

residue) subject to the positivity constraint using simulated annealing. The positivity constraint is incorporated into the error measure and the resulting unconstrained minimization problem is solved via the conjugate gradient method [18]. A *a priori* knowledge of the image and the PSF is expressed through constraint sets and a projection-based algorithm is developed to deal with the blind deconvolution problem in [19]. The performance of this group of algorithms has been demonstrated only for speckle images.

The primary difficulty in blind image restoration, or joint blur identification and image restoration, is insufficient information in the sense that this problem admits many—possibly an infinite number—of solutions. The positivity and/or size constraints are obviously useful in reducing the number of admissible solutions, but they are by no means enough. Another type of constraint, piecewise smoothness of image, has been widely accepted as an effective means for regularizing the ill conditioning of image restoration; it can be expected that the incorporation of this type of constraint will improve blind restoration quality. Though the smoothness constraint of the image (not *piecewise* smoothness) has been implicitly incorporated into the ML-based algorithms via the space-invariant AR image model and into the GCV algorithm via the space-invariant regularization, further improvement may be achieved via the incorporation of the *piecewise* smoothness of the image. This may be justified by the widely held observation that space-invariant restoration methods induce ringing artifacts around sharp intensity transitions, and by the success of the space-adaptive restoration algorithm proposed in [20], which accounted for the piecewise smoothness of the image, in significantly reducing ringing artifacts.

Unfortunately, the inclusion of space-variant models in the ML and GCV algorithms further significantly complicates these estimators. In this paper, we extend the space-adaptive regularized image restoration algorithm in [20] to address this problem. In addition to the piecewise smoothness of images, we assume that the PSF is also piecewise smooth, based on the belief that piecewise smoothness is not a stringent condition and is satisfied by many practical PSF's such as the motion blur and out-of-focus blur. The piecewise smoothness of both the image and the PSF are incorporated into the blind restoration process via space-adaptive regularization. The resulting methods attempt to minimize a cost function consisting of a restoration error measure and two regularization terms (one for the image and the other for the PSF), subject to constraints such as nonnegativity of the image and PSF. All these are done in Section II. In Section III a scale problem is first identified, which, if not properly treated, may hinder our minimization/blind restoration effort, and we then propose to solve this problem via alternating minimization. We also show that alternating minimization increases algorithmic simplicity. A pruning method is proposed to automatically estimate the PSF support in Section IV. Issues related to the selection of regularization operators, parameters, and weights for both the image and the PSF are discussed in Section V. In Section VI, the proposed method is tested on numerically

and photographically blurred images. We conclude this paper in Section VII.

II. DOUBLE REGULARIZATION

A space-invariant model for a noisy, linearly degraded image is given by [1], [2] as follows:

$$g(x) = d(x) * f(x) + n(x), \quad x = (x_1, x_2) \in \Omega \quad (1)$$

where $\Omega \subset \mathcal{R}^2$ is the support of the image, and $f(x)$, $g(x)$, $d(x)$, and $n(x)$ represent the original image, observed image, blur operator (PSF), and observation noise, respectively. The $(*)$ in (1) denotes 2-D convolution given by

$$d(x) * f(x) = \sum_{s \in \mathcal{D}} d(s) f(x - s) \quad (2)$$

where $\mathcal{D} \subset \mathcal{R}^2$ is the support of $d(s)$.

If the blur operator $d(x)$ is exactly known, a space-adaptive regularization approach to image restoration [20] seeks to minimize the following cost function:

$$L(\hat{f}) = \frac{1}{2} \sum_{x \in \Omega} w_1(x) [g(x) - d(x) * \hat{f}(x)]^2 + \frac{1}{2} \lambda \sum_{x \in \Omega} w_2(x) [c(x) * \hat{f}(x)]^2 \quad (3)$$

subject to

$$0 \leq \min \leq \hat{f}(x) \leq \max < \infty, \quad x \in \Omega. \quad (4)$$

The first term on the right-hand side of (3) represents the fidelity of the restored image $\hat{f}(x)$ with respect to the observation data $g(x)$. Direct minimization of this term would lead to excessive noise magnification due to the ill conditioning of blur operator, so a smoothness constraint (second term) is imposed. The $c(x)$ in this term is called the regularization operator and is usually a highpass filter (such as the Laplacian). The λ in (3) is the regularization parameter that controls the tradeoff between fidelity to the observation and smoothness of the restored image. Since natural images are piecewise smooth and observation noise is typically not spatially stationary, this tradeoff should be spatially adaptive according to the local properties of image and noise. Otherwise, ringing artifacts may arise around sharp intensity transitions due to excessive regularization (smoothing), while smooth areas may appear noisy due to lack of enough regularization (smoothing). The space adaptivity is achieved through the introduction of weights $w_1(x)$ and $w_2(x)$. The $w_1(x)$ is set to be large (emphasizing fidelity to the data) where the noise is small or near sharp intensity transitions, and is set to be small otherwise. The $w_2(x)$ is set to be large to emphasize smoothness near smooth areas or where noise is large and is set to be small otherwise. The constraint (4) is due to the fact that image intensity is finite and nonnegative. An iterative algorithm has been developed by minimizing the cost function (3) using steepest descent [20].

A symmetric problem to the above restoration process is the identification of the PSF $d(x)$ from the observed image $g(x)$ given that the original image $f(x)$ is exactly known.

The symmetry owes to the fact that the degradation model (1) is symmetric with respect to $d(x)$ and $f(x)$, and that the identification problem is also ill conditioned because natural images are usually lowpass. Consequently, this problem can be addressed similarly by seeking to minimize the following cost function:

$$L(\hat{d}) = \frac{1}{2} \sum_{x \in \Omega} w_1(x) [g(x) - \hat{d}(x) * f(x)]^2 + \frac{1}{2} \gamma \sum_{x \in \mathcal{D}} w_3(x) [a(x) * \hat{d}(x)]^2 \quad (5)$$

where $a(x)$ is the regularization operator, γ the regularization parameter, and $w_3(x)$ the weight, respectively. Since an imaging system normally neither absorbs nor generates energy, and the image intensities involved are nonnegative, the PSF should also satisfy

$$\sum_{x \in \mathcal{D}} \hat{d}(x) = 1 \quad (6)$$

and

$$\hat{d}(x) \geq 0, \quad x \in \mathcal{D}. \quad (7)$$

When both the original image and the PSF are unknown, the deconvolution problem is equivalent to the decomposition of the observed image g in the presence of noise. A well-known example is that any discrete signal with finite energy can be decomposed as the convolution of a minimum phase signal with an allpass system. If there is no additional information about the classes of f and d , the decomposition is not unique. In addition, the problem of ill conditioning is still present. Therefore, we need to impose conditions on f and d such that the decomposition is unique and the deconvolution is well behaved. The problem of ill conditioning can be addressed by imposing piecewise smoothness on both f and d . Although the uniqueness issue is not fully resolved, the number of acceptable decompositions can be significantly reduced by the assumption that the support of d is far smaller than that of f in addition to the constraints (6) and (7). Therefore, we propose to solve this problem by

$$\min_{\hat{d}, \hat{f}} L(\hat{d}, \hat{f}) = \frac{1}{2} \sum_{x \in \Omega} w_1(x) [g(x) - \hat{d}(x) * \hat{f}(x)]^2 + \frac{1}{2} \lambda \sum_{x \in \Omega} w_2(x) [c(x) * \hat{f}(x)]^2 + \frac{1}{2} \gamma \sum_{x \in \mathcal{D}} w_3(x) [a(x) * \hat{d}(x)]^2$$

Subject to (4), (6) and (7). (8)

When this function is practically minimized, we have to assume a particular support for d . This is an important condition imposed on d , which reduces the number of possible alternatives in the decomposition of the image and the blur. It is clear that (8) is not convex and, hence, has the problems associated with local minima. However, this function is far simpler than both ML and GCV cost functions in that it is quadratic with respect to $\hat{d}$ when $\hat{f}$ is fixed and with respect

to $\hat{f}$ when $\hat{d}$ is fixed, so efficient algorithms can be developed for its minimization.

III. ALTERNATING MINIMIZATION

Many optimization procedures can be employed to minimize (8). Here we follow the strategy [1] of employing iterative unconstrained optimization procedures (such as steepest descent and conjugate gradient) to minimize $L(\hat{d}, \hat{f})$ and impose the constraints (4), (6) and (7) on the solution after each iteration. Most of the unconstrained optimization procedures use the gradient of $L(\hat{d}, \hat{f})$, which we give as follows:

$$p(x) = \frac{\partial L(\hat{d}, \hat{f})}{\partial \hat{f}(x)} = -\{w_1(x)[g(x) - \hat{d}(x) * \hat{f}(x)]\} * \hat{d}(-x) + \lambda\{w_2(x)[c(x) * \hat{f}(x)]\} * c(-x) \quad (9)$$

and

$$q(x) = \frac{\partial L(\hat{d}, \hat{f})}{\partial \hat{d}(x)} = -\{w_1(x)[g(x) - \hat{d}(x) * \hat{f}(x)]\} * \hat{f}(-x) + \gamma\{w_3(x)[a(x) * \hat{d}(x)]\} * a(-x). \quad (10)$$

Note that, due to the fact that the support of the PSF is significantly smaller than that of the image, the computational load involved in the evaluation of (10) is less than that in (9). So it is expected that the computational load for the blur-identification step will be less than that for the image-restoration step.

In the following, we first illustrate that there is a scale problem in $L(\hat{d}, \hat{f})$ which, if not carefully treated, may seriously undermine our minimization/restoration effort [21]. Then we discuss ways to alleviate this problem, and propose a solution by alternating minimization. We also show that, under the framework of alternating minimization, each of the two minimization steps seeks to optimize a quadratic cost function. Hence, the resulting algorithms are simple and efficient.

A. Scale Problem

The scale problem can be best illustrated when we consider minimizing the cost function (8) via steepest descent. The steepest descent procedure minimizes (8) by moving the estimates iteratively in the negative gradient direction, as follows:

$$\hat{f}^{k+1}(x) = \hat{f}^k(x) - \alpha p^k(x) \quad (11)$$

and

$$\hat{d}^{k+1}(x) = \hat{d}^k(x) - \alpha q^k(x) \quad (12)$$

where α is the step size. A large step size α is needed for fast convergence, but too large an α may destabilize the iterative algorithm. For the problem at hand, we show that the scale problem arises because a very small step size is necessary for the stability of (11) while a much larger one is needed for the convergence of (12).

Let us illustrate this in detail. The first term of (10) is $\sum_{x \in \Omega} \hat{f}(x)$ times larger than that of (9) due to (6). Since the regularization parameters λ and γ control the relative values of the first terms and the second terms of (9) and (10), respectively, $q(x)$ is $\sum_{x \in \Omega} \hat{f}(x)$ times larger than $p(x)$. On the other hand, since the dynamic range of the image, say $[0, 255]$, is $\max_{x \in \Omega} \hat{f}(x)$ times larger than the PSF dynamic range $[0, 1]$, the amount of update needed by (12) should be $\max_{x \in \Omega} \hat{f}(x)$ times smaller than that of (11). Consequently, if α is a step size that gives (12) a speed of convergence v_d , the speed of convergence for (11) is approximately

$$v_f \approx \frac{v_d}{\max_{x \in \Omega} \hat{f}(x) \sum_{x \in \Omega} \hat{f}(x)}. \quad (13)$$

For an image of 256×256 with 256 grey scales, this is a difference of at least 10^9 . The problem is that, if α is set to be large enough for fast convergence of (11), (12) is not stable; on the other hand, if α is set to be small enough to guarantee the stability of (12), the convergence of (11) is too slow to have any practical value.

This scale problem can be alleviated following a scaling procedure as described in [21]. For the problem at hand, we elect to scale down the PSF by a factor of T through the definition of a new variable, as follows:

$$\bar{d}(x) = T\hat{d}(x). \quad (14)$$

The problem of finding $\hat{f}$ and $\hat{d}$ to minimize $L(\hat{d}, \hat{f})$ is equivalent to that of finding $\hat{f}$ and $\bar{d}$ to minimize $H(\bar{d}, \hat{f}) = L(T\bar{d}, \hat{f}) = L(\hat{d}, \hat{f})$. Since $\frac{\partial H(\bar{d}, \hat{f})}{\partial \bar{d}(x)} = T \frac{\partial L(\hat{d}, \hat{f})}{\partial \hat{d}(x)}$, The steepest descent of $H(\bar{d}, \hat{f})$ with respect to $\bar{d}$ is

$$\bar{d}^{k+1}(x) = \bar{d}^k(x) - \alpha \frac{\partial H(\bar{d}, \hat{f})}{\partial \bar{d}(x)} \bigg|_{\bar{d}^k} (x) = \bar{d}^k(x) - \alpha T \frac{\partial L(\hat{d}, \hat{f})}{\partial \hat{d}(x)}. \quad (15)$$

In the original variable $\hat{d}$, this is

$$\hat{d}^{k+1}(x) = \hat{d}^k(x) - \alpha T^2 q^k(x). \quad (16)$$

Intuitively, the scale factor T should be selected such that the convergence speeds of (11) and the scaled (16) are approximately equal. In view of (13), one such scale factor should be

$$T^2 \approx \frac{1}{\max_{x \in \Omega} \hat{f}(x) \sum_{x \in \Omega} \hat{f}(x)} \quad (17)$$

although a more involved derivation gives (see Appendix A) the following:

$$T^2 \approx \frac{\sum_{x \in \mathcal{D}} \hat{d}^2(x) + \lambda \sum_{x \in \mathcal{C}} c^2(x)}{\sum_{x \in \Omega} \hat{f}^2(x) + \gamma \sum_{x \in \mathcal{A}} a^2(x)} \quad (18)$$

where $\mathcal{C}$ and $\mathcal{A}$ are the supports of $c(x)$ and $a(x)$, respectively. In practice, (17) and (18) are approximately equal.

Finally, we note that the scale problem is not limited to the steepest descent method; rather, it is a problem inherent to the eigenvalue structure of the Hessian matrix of the cost function $L(\hat{d}, \hat{f})$ (the condition number is too large; see [21]). Therefore, other optimization procedures may also suffer from the same problem if the scale problem is not carefully treated.

B. Alternating Minimization

In this subsection we propose to avoid the scale problem by following the idea of coordinate descent [21], and to minimize (8) by descending with respect to the PSF and the image, separately and alternately. We follow the steps listed below.

- Fix $\hat{f}(x)$, $x \in \Omega$, descend (8) with respect to $\hat{d}(x)$, $x \in \mathcal{D}$.
- Fix $\hat{d}(x)$, $x \in \mathcal{D}$, descend (8) with respect to $\hat{f}(x)$, $x \in \Omega$.

Let

$$L(\hat{d}, \hat{f} \mid \hat{d}) = L(\hat{d}, \hat{f}) \text{ with } \hat{d} \text{ fixed} \quad (19)$$

and

$$L(\hat{d}, \hat{f} \mid \hat{f}) = L(\hat{d}, \hat{f}) \text{ with } \hat{f} \text{ fixed}. \quad (20)$$

Then the above idea can be developed into the following iterative algorithm.

- Initialization:

$$\hat{f}^0(x) = g(x) \quad \text{and} \quad \hat{d}^0(x) = \text{random numbers} \quad (21)$$

- n th iteration:

$$\text{Descend}_{\hat{f}^n} L(\hat{d}^n, \hat{f}^n \mid \hat{d}^n) \quad (22)$$

- $(n+1)$ th iteration:

$$\text{Descend}_{\hat{d}^n} L(\hat{d}^n, \hat{f}^n \mid \hat{f}^n). \quad (23)$$

- Stop when converged.

The global convergence of the algorithm to one of the local minima of the cost function $L(\hat{d}, \hat{f})$ can be established by noting the fact that both (22) and (23) always decrease the cost function $L(\hat{d}, \hat{f})$ [21]. The order of (22) and (23) can be exchanged, leading to different iteration processes. In some cases, the order of (22)–(23) may lead to fewer iterations for convergence, and for some other cases the reversed order may lead to fewer iterations. It is worthwhile to try the other order if convergence is slow.

Since the descents with respect to the PSF and the image are separate, they can have different optimal step sizes tailored to each of them; the scale problem described in the preceding subsection then no longer exists. In addition, since both $L(\hat{d}, \hat{f} \mid \hat{d})$ and $L(\hat{d}, \hat{f} \mid \hat{f})$ are quadratic, (22) and (23) can be efficiently implemented by simple optimization methods such as steepest descent and conjugate gradient.

Let us now consider the implementation of the two descent steps (22) and (23) by steepest descent. By the definition of partial derivative, the formula of (9) and (10) still hold for $L(\hat{d}^n, \hat{f}^n \mid \hat{d}^n)$ and $L(\hat{d}^n, \hat{f}^n \mid \hat{f}^n)$, so that the two descent steps (22) and (23) may be implemented by (11) and (12), respectively. Since both $L(\hat{d}^n, \hat{f}^n \mid \hat{d}^n)$ and $L(\hat{d}^n, \hat{f}^n \mid \hat{f}^n)$ are quadratic, the optimal step sizes are readily available [21], as follows:

$$\alpha^n = \frac{\sum_{x \in \Omega} p^2(x)}{\sum_{x \in \Omega} \left\{ w_1(x) [\hat{d}(x) * p(x)]^2 + \lambda w_2(x) [c(x) * p(x)]^2 \right\}} \quad (24)$$

for the image and

$$\alpha^n = \frac{\sum_{x \in \Omega} q^2(x)}{\sum_{x \in \Omega} w_1(x)[\hat{f}(x) * q(x)]^2 + \gamma \sum_{x \in \mathcal{D}} w_3(x)[a(x) * q(x)]^2} \quad (25)$$

for the blur, respectively.

Limited to the linear convergence rate of steepest descent, the above algorithm might be inadequate to handle large images and/or PSF's, so we present the following implementation based on the conjugate gradient method. The conjugate gradient method has super-linear rate of convergence and can reach the minimum of a quadratic cost function in a finite number of iterations. Yet, its complexity is only slightly more than that of steepest descent. Since the conjugate gradient method itself consists of a finite number of descent steps, our alternating minimization algorithm is then a nested iterative algorithm. For notational simplification, in the following we drop the superscript n , which is the count for the outer iterations in (22) and (23), and use the superscript k to indicate the inner conjugate gradient iterations.

For the descent of (22), let $u(x)$ be the component at x of the conjugate vector of $L(\hat{d}^n, \hat{f}^n | \hat{d}^n)$; then the conjugate gradient descent is given [21] as follows.

- Initialize the conjugate vector:

$$u^0(x) = -p^0. \quad (26)$$

- Step size for updating the image:

$$\alpha^k = \frac{\sum_{x \in \Omega} [p^k(x)]^2}{\sum_{x \in \Omega} \left\{ w_1(x)[\hat{d}^k(x) * u^k(x)]^2 + \lambda w_2(x)[c(x) * u^k(x)]^2 \right\}}. \quad (27)$$

- Update the image:

$$\hat{f}^{k+1}(x) = \hat{f}^k(x) + \alpha^k u^k(x). \quad (28)$$

- Step size for updating the conjugate vector:

$$\beta^k = \frac{\sum_{x \in \Omega} [p^{k+1}(x)]^2}{\sum_{x \in \Omega} [p^k(x)]^2}. \quad (29)$$

- Update the conjugate vector:

$$u^{k+1}(x) = -p^{k+1}(x) + \beta^k u^k(x). \quad (30)$$

For an image of $M \times N$ pixels, the above conjugate method will converge to the minimum of $L(\hat{d}^n, \hat{f}^n | \hat{d}^n)$ in MN steps. Since this is an unaffordably large number, we can simply stop it after m steps, where $m \ll MN$. This is related to the partial conjugate gradient method [21]. The proper selection of m should be such that it is larger than the number of the major eigenvalues of the Hessian matrix of $L(\hat{d}, \hat{f} | \hat{d})$ to guarantee fast convergence [21]. Appendix B will show that the eigenvalues of the Hessian matrix of $L(\hat{d}, \hat{f} | \hat{d})$ with uniform weights are given by

$$\sigma(u) = |\hat{D}(u)|^2 + \lambda |C(u)|^2 \quad (31)$$

where $u = (u_1, u_2)$ is the 2-D space frequency, and $\hat{D}(u)$ and $C(u)$ are the 2-D DFT of $\hat{d}(x)$ and $c(x)$, respectively. Since PSF's, such as linear motion blur and out-of-focus blur, are usually lowpass, only a small number of the eigenvalues given by (31) (at low frequencies) are large. Consequently, a very small m often suffices.

For the descent of (23), let $v(x)$ be the component at x of the conjugate vector of $L(\hat{d}^n, \hat{f}^n | \hat{f}^n)$; the conjugate gradient descent is given [21] by the following.

- Initialize the conjugate vector:

$$v^0(x) = -q^0. \quad (32)$$

- Step size for updating the PSF:

$$\alpha^k = \frac{\sum_{x \in \Omega} [q^k(x)]^2}{\sum_{x \in \Omega} w_1(x)[v^k(x) * \hat{f}^k(x)]^2 + \gamma \sum_{x \in \mathcal{D}} w_3(x)[a(x) * v^k(x)]^2}. \quad (33)$$

- Update the PSF:

$$\hat{d}^{k+1}(x) = \hat{d}^k(x) + \alpha^k v^k(x). \quad (34)$$

- Step size for updating the conjugate vector:

$$\beta^k = \frac{\sum_{x \in \mathcal{D}} [q^{k+1}(x)]^2}{\sum_{x \in \mathcal{D}} [q^k(x)]^2}. \quad (35)$$

- Update the conjugate vector:

$$v^{k+1}(x) = -q^{k+1}(x) + \beta^k v^k(x). \quad (36)$$

Similar to the image descent, if the support of the PSF is large, the above conjugate steps should be stopped after n steps. Appendix B shows that the eigenvalues of the Hessian matrix of $L(\hat{d}, \hat{f} | \hat{f})$ with uniform weights are given by

$$\rho(v) = |\hat{F}(v)|^2 + \lambda |A(v)|^2 \quad (37)$$

where $v = (v_1, v_2)$ is the 2-D space frequency and $\hat{F}(v)$ and $A(v)$ are the 2-D DFT of $\hat{f}(x)$ and $a(x)$, respectively. Since natural images have only a few very large frequency components and the others are very close to zero, a very small n will often suffice to guarantee fast convergence.

IV. ESTIMATION OF PSF SUPPORT

Support of the PSF has to be specified before the proposed methods can be used to identify the PSF. If specified PSF support does not cover actual PSF support, the PSF will not be identified correctly. In order to ensure the actual PSF is within the specified support, we can pessimistically assume a large support. Unfortunately, this will significantly increase computational load. Therefore, we propose to prune the boundary of the assumed PSF support at each iteration until it reduces (at least approximately) to the original PSF support. A boundary point $x \in \partial \mathcal{D}$ is pruned if it satisfies

$$\hat{d}(x) \leq T, \quad x \in \partial \mathcal{D} \quad (38)$$

where T is a threshold and $\partial \mathcal{D}$ is the boundary of $\mathcal{D}$. Due to the nonnegativeness of the PSF, the threshold should be positive.

Unfortunately, pruning in this way would lead to irregular PSF supports, which might not be convenient for implementation on computers. We would like to prune in such a way that the PSF support is always rectangular. a side $\partial\bar{\mathcal{D}}$ of the rectangular PSF support $\partial\mathcal{D}$ is to be pruned if

$$\hat{d}(x) \leq T, \quad x \in \partial\bar{\mathcal{D}}. \quad (39)$$

V. REGULARIZATION ISSUES

A regularization scheme consists of three elements: regularization operator, parameter, and weight. These elements are critical to the success of the whole regularized blur identification and image restoration process. Selection of a proper regularization scheme for images has been discussed in [20]. Regularization of the PSF is a new topic and its properties are somewhat different than that of the image. This section proposes some modifications to image regularization and discusses regularization of the PSF.

The regularization operator for an image is in general a highpass filter, and the usual selection is the discretization of the Laplacian $\nabla^2 f(x_1, x_2) = \frac{\partial^2 f}{\partial x_1^2} + \frac{\partial^2 f}{\partial x_2^2}$, which usually has a support of 3×3 . Due to the convolution in the second term of (9), $5 \times 5 = 25$ pixels are involved in the regularization operation for each pixel. The image intensities of these 25 pixels are significantly different near an edge, so the smoothing (regularization) operation for them is unnecessary and may lead to a smeared edge or ringing artifacts. This large support is particularly irrelevant to regularization of the PSF due to the usually small support of the PSF. Therefore, regularization operators with smaller supports should be used. One possible choice is to use the discretization of the gradient sum $\frac{\partial f}{\partial x_1} + \frac{\partial f}{\partial x_2}$, as follows:

$$G = \begin{bmatrix} 2 & -1 \\ -1 & 0 \end{bmatrix}. \quad (40)$$

When the PSF is 1-D, (40) reduces to

$$G = [1 \quad -1] \quad (41)$$

and

$$G = \begin{bmatrix} 1 \\ -1 \end{bmatrix} \quad (42)$$

corresponding to $\frac{\partial f}{\partial x_1}$ and $\frac{\partial f}{\partial x_2}$, respectively.

Several papers have addressed the problem of estimating the optimal regularization parameter for the restoration problem

[22], [23], [14], [24], [25]. These methods can be employed to estimate the regularization parameter for the image, but their complexities are incompatible with the simplicity of the proposed algorithms, so we will give a new one here. Estimation of the PSF regularization parameter is a new problem, but the methods developed for estimating image regularization parameters can be borrowed due to the symmetry of the image and the PSF in our formulation. Similar to the case for the image, we also give a new and simple method. The basic idea behind this method is that the partial derivatives (9) and (10) should be zero at convergence. So we set them to zero at each iteration to obtain two systems of linear equations as follows:

$$\begin{aligned} & \{w_2(x)[c(x) * \hat{f}(x)]\} * c(-x) \quad \lambda \\ & = \{w_1(x)[g(x) - \hat{d}(x) * \hat{f}(x)]\} * \hat{d}(-x), \quad x \in \Omega \end{aligned} \quad (43)$$

$$\begin{aligned} & \{w_3(x)[a(x) * \hat{d}(x)]\} * a(-x) \quad \gamma \\ & = \{w_1(x)[g(x) - \hat{d}(x) * \hat{f}(x)]\} * \hat{f}(-x), \quad x \in \mathcal{D}. \end{aligned} \quad (44)$$

The above are solved by least squares to give (45) and (46), shown at the bottom of the page.

The above methods give order-of-magnitude estimates for the regularization parameters. These estimates often appear to under-regularized estimates of the image and the PSF. Hence, adjustments within the estimated orders of magnitude may have to be made for slight improvement in image quality. In fact, the regularization parameters estimated by the methods in [22], [23], [14], [24], and [25] may differ by orders of magnitude. Unfortunately, there is no way aside from visual inspection to decide rigorously which estimate is the best, due to the lack of a widely accepted quantitative measure of image quality. However, our simulations have shown that blind restoration quality is usually not sensitive to regularization parameters. Note that, due to the scale problem as discussed in Section III-B, the two regularization parameters should have the following approximate relation:

$$\frac{\gamma}{\lambda} \approx \sum_{x \in \Omega} \hat{f}(x) \max_{x \in \Omega} \hat{f}(x) \quad (47)$$

if the relative values of the first and the second terms of (10) are to follow a similar pattern as that held by the first and the second terms of (9). Also note that the estimation of regularization parameters should be done only for the outer iterations (22) and (23) if the conjugate gradient descents are employed. Otherwise, the conjugacy of the descent vectors would be destroyed.

$$\lambda = \frac{\sum_{x \in \Omega} \left\{ \{w_2(x)[c(x) * \hat{f}(x)]\} * c(-x) \right\} \left\{ \{w_1(x)[g(x) - \hat{d}(x) * \hat{f}(x)]\} * \hat{d}(-x) \right\}}{\sum_{x \in \Omega} \left\{ \{w_2(x)[c(x) * \hat{f}(x)]\} * c(-x) \right\}^2} \quad (45)$$

$$\gamma = \frac{\sum_{x \in \mathcal{D}} \left\{ \{w_3(x)[a(x) * \hat{d}(x)]\} * a(-x) \right\} \left\{ \{w_1(x)[g(x) - \hat{d}(x) * \hat{f}(x)]\} * \hat{f}(-x) \right\}}{\sum_{x \in \mathcal{D}} \left\{ \{w_3(x)[a(x) * \hat{d}(x)]\} * a(-x) \right\}^2} \quad (46)$$

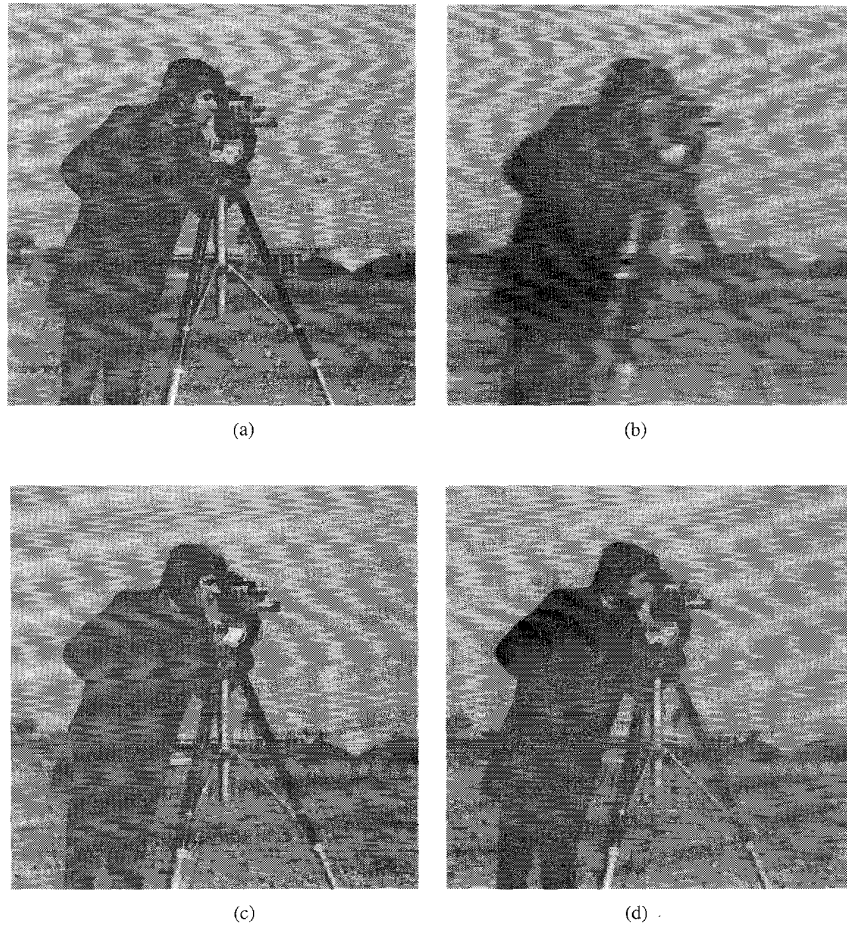


Fig. 1. Restoration of the Cameraman image degraded by 1×10 linear motion blur and white Gaussian noise at 30 dB SNR. (a) Original Cameraman image. (b) Degraded image. (c) Restored image when the PSF is not known. (d) Restored image when the PSF is exactly known.

The image weights $w_1(x)$ and $w_2(x)$ have been proposed in [20] to be selected as

$$w_1(x) = \begin{cases} 1, & \text{if data at } x \text{ is reliable} \\ 0, & \text{otherwise} \end{cases} \quad x \in \Omega \quad (48)$$

and

$$w_2(x) = \frac{1}{1 + \alpha_2 \text{local_var}(x)}, \quad x \in \Omega \quad (49)$$

respectively, where the $\text{local_var}(x)$ is the local variance of the observed image $g(x)$ at x in a $P \times Q$ window. α_2 is a parameter to be adjusted for a particular application. Our simulation has shown that the restoration performance is not sensitive to its value, though we suggest that α_2 be selected such that

$$\alpha_2 \max_{x \in \Omega} \text{local_var}(x) \approx 1000. \quad (50)$$

As to the PSF weight $w_3(x)$, we should first note that the initial PSF are random numbers that do not provide any information about the space activity of the PSF, so $w_3(x)$ has to be obtained iteratively, as follows:

$$w_3^0(x) = 1.0, \quad x \in \mathcal{D} \quad (51)$$

$$w_3^n(x) = \frac{1}{1 + \alpha_3 \text{local_var}^n(x)}, \quad x \in \mathcal{D} \quad (52)$$

where $\text{local_var}^n(x)$ is the local variance at x of the estimated PSF $\hat{d}^n(x)$. Due to the small support of the PSF, the window for the local variance has to be 3×3 . Once again, our simulation has shown that restoration quality is not sensitive to its value, though we suggest a guideline similar to that for α_2 :

$$\alpha_3 \max_{x \in \mathcal{D}} \text{local_var}(x) \approx 1000. \quad (53)$$

Since the local variance $\text{local_var}^n(x)$ is very small due to (6), (53) usually leads to a very large α_3 . Note that when conjugate gradient descent is used, this weight update should always be outside the conjugate gradient descent circle in order not to disrupt the conjugacy of the descent vectors.

VI. NUMERICAL EXPERIMENTS

The proposed method has been tested on several images and PSF's, the results of some of which are presented here. The alternating minimization algorithm based on conjugate gradient method (discussed in Section III-B) is used to run all the experiments. For the image descents, the number of

conjugate gradient steps at each iteration is set to 10. For the PSF descents, full conjugate gradient will be used if the PSF support is no larger than ten, otherwise the number of conjugate descent will be set to ten. For the image and PSF weights (49) and (52), we choose $\alpha_2 = 0.5$ and $\alpha_3 = 100\,000$ by following the guidelines of (50) and (53). Fine tuning the weights for a particular application may slightly reduce ringing artifacts, but we use the same set of weights for the three applications presented here, which include two different images and three different PSF's in order to show that the restoration quality is not sensitive to the weights. The operator in (40) is used to regularize images and 2-D PSF's, and that in either (41) or (42) is used to regularize 1-D PSF's depending on which one is appropriate. Image restoration performance is measured by the SNR improvement [1], [2] that follows:

$$\Delta_{\text{SNR}} = 10 \log_{10} \frac{\|f - g\|}{\|f - \hat{f}\|}. \quad (54)$$

A. Identification of a 1-D PSF

The Cameraman image shown in Fig. 1(a) is degraded by a 1×10 linear motion blur with white Gaussian noise added at 30 dB signal-to-noise ratio (SNR) defined by

$$\text{SNR} = 10 \log_{10} \frac{\text{Variance of the blurred image}}{\text{Variance of the noise}}. \quad (55)$$

The degraded image is shown in Fig. 1(b). Since we do not know the support of the underlying PSF, we make a conservative guess about it, for example, set it to 1×15 . The PSF is then initialized to 15 random numbers. We first assume that the original image is exactly known and use the alternating minimization algorithm with the image descent step (22) disabled to identify the PSF. The process of PSF convergence is shown in Fig. 2(a). It takes only 4 iterations to converge. Now we use the full alternating minimization algorithm to identify the underlying PSF under the realistic assumption that we do not know the original image. The alternating minimization algorithm is run in the order of (22) $\rightarrow$ (23) $\rightarrow$ (22) $\rightarrow \dots$. The process of PSF convergence is shown in Fig. 2(b). Since the original image is unknown, we need to estimate both the original image and the PSF simultaneously, so the convergence rate is reduced. However, this lack of information does not prevent us from getting good estimates about the original PSF. Fig. 2(c) shows how close the two PSF estimates (one with full knowledge of the original image and the other one without this knowledge) are to each other as well as to the original PSF. Fig. 1(c) shows the restored image at the end of the joint image and PSF estimation iteration. For comparison, the restored image when the original PSF is exactly known is also shown in Fig. 1(d). For all the above experiments, the regularization parameters are 0.1 and 10^9 for the image and the PSF, respectively. The rather large regularization parameter for the PSF with respect to that for the image is due to the scale property (47). The image SNR improvement is 9.7763 dB when the underlying PSF is exactly known *a priori*, and it reduces to 9.1813 dB when the PSF is unknown.

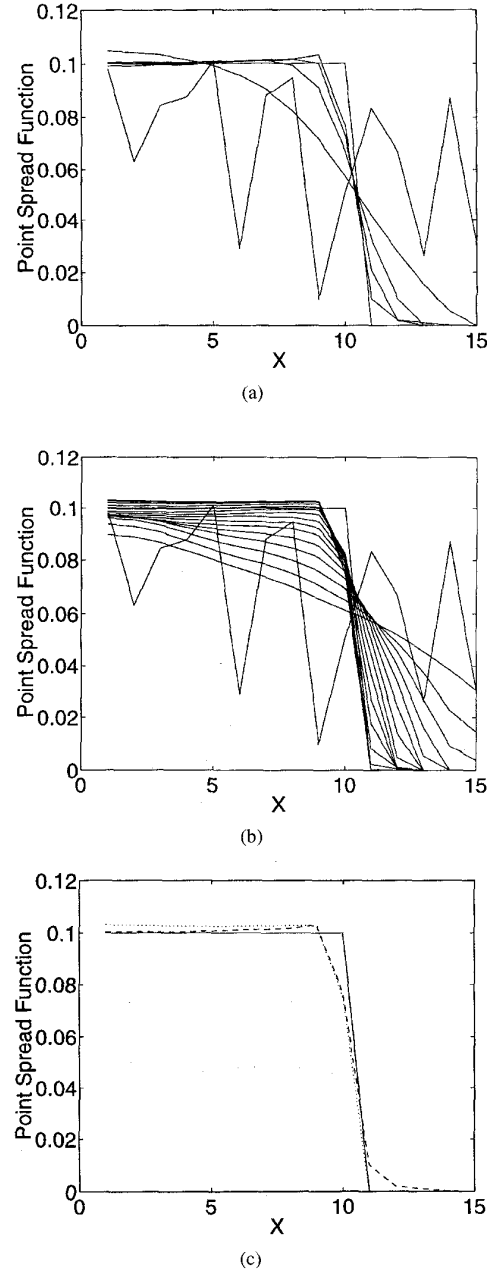


Fig. 2. Identification of 1×10 linear motion blur while its support is erroneously assumed to be 1×15 . (a) When the original image is exactly known. The zigzag curve is the initial PSF consisting of random numbers and the one with the sharp corner is the real PSF. (b) When the original image is unknown, more iterations are taken for convergence. (c) Solid line: the real PSF. Dashed line: identified PSF assuming that the original image is known. Dotted line: identified PSF assuming that the original image is unknown.

B. Identification of a 2-D PSF

The experiment in [11] is repeated here. The Cameraman image in Fig. 1(a) is degraded by 3×3 uniform blur, shown in Fig. 4(a), with Gaussian noise added at 30 dB SNR. The degraded image is shown in Fig. 3(a). We assume that both the original image and the PSF are unknown, and that we are to estimate both of them solely from the degraded image.

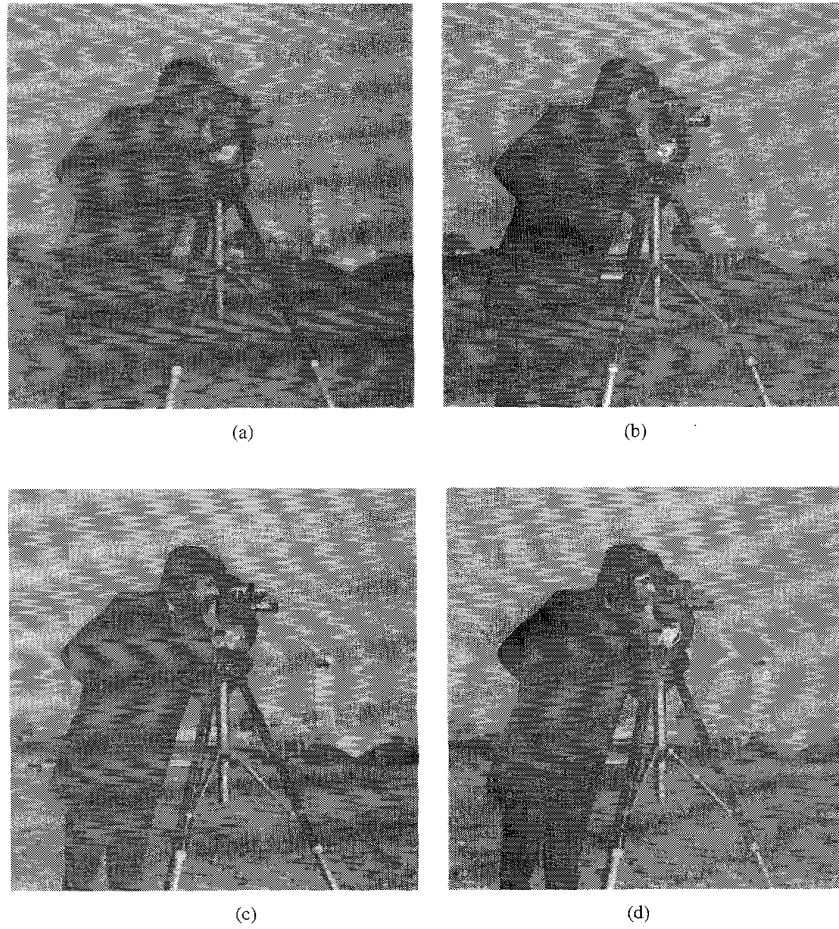


Fig. 3. (a) Cameraman image degraded by 3×3 uniform blur and white Gaussian noise at 30 dB SNR; (b) restored image while the underlying PSF is identified under the condition that its support is fixed at 7×7 ; (c) restored image while the underlying PSF is identified under the condition that its support is automatically pruned at each iteration starting from 7×7 (d) restored image under the condition that the underlying PSF is exactly known.

Since we have no idea what the original PSF may look like, we have to assume a large support for it. In this simulation, we assume the support to be 7×7 , which means that we have 48 free coefficients to estimate. We then initialize the PSF to 49 random numbers and run the alternating algorithm in the order of $(23) \rightarrow (22) \rightarrow (23) \rightarrow \dots$. After convergence, the identified PSF is shown in Fig. 4(b) and the restored image is shown in Fig. 3(b).

The pruning mechanism proposed in Section IV is then added to the iterative algorithm. The PSF is again initialized to 49 random numbers and the alternating algorithm is run. Due to (6), the pruning threshold is set as $T = \min\{1/\text{sizeofcurrentPSFsupport}, 0.01\}$. Fig. 5 shows the process of the PSF support is pruned. It takes only seven iterations to prune the PSF support from the initial 7×7 to 3×3 , which is the real PSF support. After convergence, the identified PSF is

$$\hat{d} = \begin{bmatrix} 0.1114 & 0.1113 & 0.1106 \\ 0.1116 & 0.1113 & 0.1105 \\ 0.1118 & 0.1113 & 0.1102 \end{bmatrix}$$

and is shown in Fig. 4(c). The restored image is shown in Fig. 3(c). The image restored under the condition that the original PSF is exactly known is shown in Fig. 3(d). For comparison, the PSF identified by EM algorithm is repeated here [11]:

$$d = \begin{bmatrix} 0.102 & 0.113 & 0.114 \\ 0.106 & 0.131 & 0.106 \\ 0.114 & 0.113 & 0.102 \end{bmatrix}.$$

Note that this is obtained under the assumption that we know that the PSF is symmetric and its support is 3×3 . This assumption reduces the free PSF coefficients to four. It has been observed [11] that “blurs with larger support size (i.e., with more free PSF coefficients) cannot satisfactorily be identified without using additional knowledge about the structure of the PSF”.

For all the above experiments, the regularization parameters are 0.5 and 10^8 for the image and the PSF, respectively. The image SNR improvement is 5.8794 dB when the underlying PSF is exactly known *a priori*. This reduces to 5.6327 dB when the PSF is unknown and is identified with the assumption that

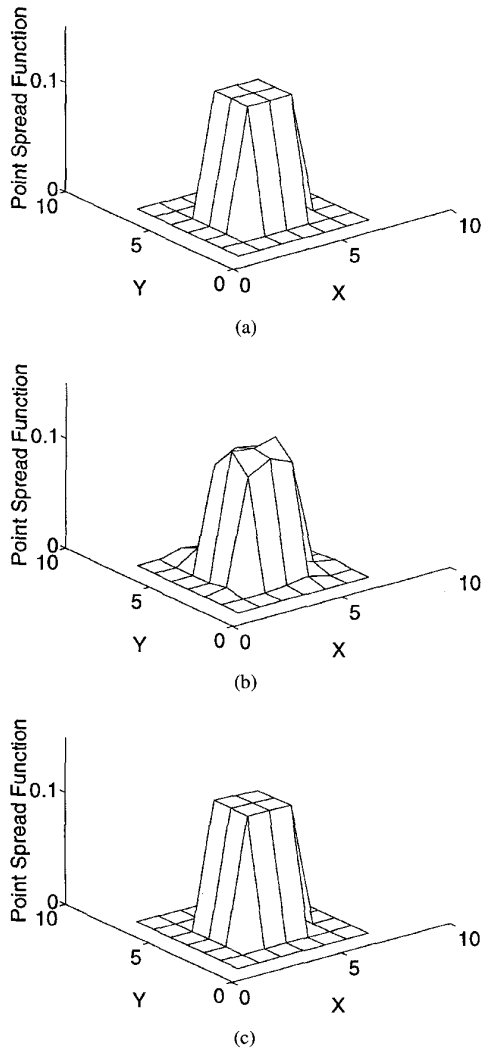


Fig. 4. Identification of 3×3 uniform blur while its support was erroneously assumed to be 7×7 . (a) Original 3×3 uniform blur embedded in a 7×7 support for viewing. (b) identified blur, its support was fixed at 7×7 during identification process. (c) Identified blur embedded in a 7×7 support for viewing. The PSF support was initially 7×7 and was automatically pruned at each iteration. This support finally reaches 3×3 after convergence.

its support is 7×7 . When the PSF support is automatically pruned starting from 7×7 , a better image SNR improvement is obtained: 5.8540 dB, due to the fact that the true PSF support is correctly estimated by the pruning algorithm.

C. A Photographically Blurred Image

The photographically blurred image in Fig. 6(a) is used to test the alternating minimization algorithm. Let us first assume this image was blurred by a PSF of 7×7 support and run the alternating minimization algorithm. After 80 iterations, the identified PSF is shown in Fig. 7(a) and the restored image is shown in Fig. 6(b). Fig. 7(a) shows that the effective support of the underlying PSF would be 3×3 , so we set the PSF support to 3×3 and run the alternating minimization algorithm again. After 20 iterations, the

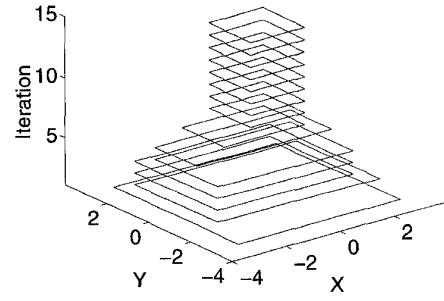


Fig. 5. Pruning process of the PSF support. The bottom square is the originally assumed 7×7 support. It is automatically pruned to 3×3 at 7th iteration and then settles down there. The correct PSF support is 3×3 .

identified PSF is shown in Fig. 7(b) and the restored image is shown in Fig. 6(c). Though the newly identified PSF in Fig. 7(b) is similar to that in Fig. 7(a), the newly restored image in Fig. 6(c) is significantly less blurred than that in Fig. 6(b). This indicates that the small PSF coefficients are also important to restoration quality. For both the above two experiments, the regularization parameters used are 0.05 and 10^7 for image and PSF, respectively.

VII. CONCLUSION

This paper presented a space-adaptive regularization approach to blind image restoration, which uses the piecewise smoothness of both the image and the PSF to effectively cope with the problem of insufficient information. The approach differs from the series of methods that followed the iterative loop algorithm of [15] by the additional piecewise smoothness constraints imposed on the image and the PSF. The addition of these constraints is shown to be very effective in increasing blind restoration quality. One differentiating feature of the method presented here from the ML- and GCV-based methods is its space-adaptivity, which enables it to utilize the property of piecewise smoothness for both the image and the PSF. Hence, we have observed improved blind restoration quality. The simplicity of the cost function and the derived algorithms is another contrasting feature in comparison with both the ML- and GCV-based methods.

The algorithm requires the selection of four parameters. For the regularization parameters, λ and γ , order-of-magnitude estimates are given in (45) and (46). The two parameters for the space-adaptive weights in (49) and (52) are α_2 and α_3 . Our choices for these two parameters were discussed following the above two equations. Clearly, choosing $\alpha_2 = \alpha_3 = 0$ leads to space-invariant blind restoration. We have found, however, that the quality of blind restoration is not sensitive to the actual values of these parameters, as long as they satisfy general order-of-magnitude guidelines given in the paper. Nevertheless, these parameters can be adjusted to achieve slight improvement, if desired. Finally, it is clear that the proposed method suffers from shortcomings—with respect to local minima and nonuniqueness of the solution—that are similar to other blind restoration methods. However, this has not been a problem in our simulations.



(a)



(b)



(c)

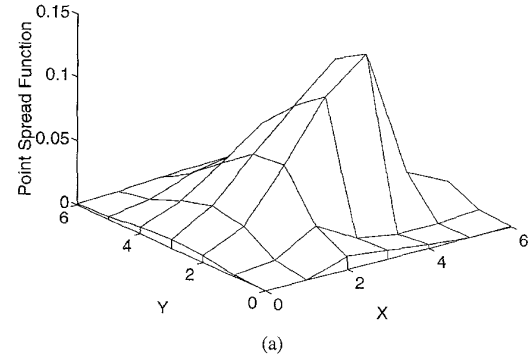
Fig. 6. Restoration of a photographically blurred image. (a) Photographically blurred image. (b) Restored image when the underlying PSF is assumed to have a 7×7 support. (c) Restored image when the underlying PSF is assumed to have a 3×3 support.

APPENDIX

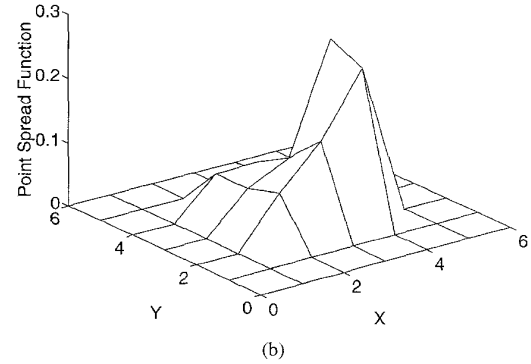
A. Scale Factors

The goal of selecting a scale factor is to make [21]

$$\frac{\partial^2 H(\bar{d}, \hat{f})}{\partial \bar{d}^2} \approx \frac{\partial^2 H(\bar{d}, \hat{f})}{\partial \hat{f}^2}. \quad (56)$$



(a)



(b)

Fig. 7. Blur identification for a photographically blurred image. (a) Identified PSF when its support was assumed to be 7×7 . (b) Identified PSF when its support was assumed to be 3×3 , but embedded in a 7×7 support for viewing.

Since

$$\frac{\partial^2 H(\bar{d}, \hat{f})}{\partial \bar{d}^2} = T^2 \frac{\partial^2 L(\hat{d}, \hat{f})}{\partial \hat{d}^2} \quad \text{and} \quad \frac{\partial^2 H(\bar{d}, \hat{f})}{\partial \hat{f}^2} = \frac{\partial^2 L(\hat{d}, \hat{f})}{\partial \hat{f}^2} \quad (57)$$

and

$$\frac{\partial^2 L(\hat{d}, \hat{f})}{\partial \hat{d}^2(x)} = w_1(x) * \hat{f}^2(-x) + \gamma w_3(x) * a^2(-x), \quad x \in \mathcal{D} \quad (58)$$

$$\frac{\partial^2 L(\hat{d}, \hat{f})}{\partial \hat{f}^2(y)} = w_1(y) * \hat{d}^2(-y) + \lambda w_2(y) * c^2(-y), \quad y \in \Omega \quad (59)$$

from (10) and (9), the scale factor can be obtained as

$$\begin{aligned} T^2 &\approx \frac{\frac{\partial^2 L(\hat{d}, \hat{f})}{\partial \hat{f}^2}}{\frac{\partial^2 L(\hat{d}, \hat{f})}{\partial \hat{d}^2}} \\ &= \frac{w_1(x) * \hat{d}^2(-x) + \lambda w_2(x) * c^2(-x)}{w_1(y) * \hat{f}^2(-y) + \gamma w_3(y) * a^2(-y)}, \quad x \in \mathcal{D}, y \in \Omega. \end{aligned} \quad (60)$$

This scale factor changes with space coordinates x and y . In order to alleviate this problem, let us set $w_1(x) = 1, x \in \Omega$, $w_2(x) = 1, x \in \Omega$, and $w_3(x) = 1, x \in \mathcal{D}$. Then (60) becomes (18).

B. Eigenvalues of Hessian Matrices

If the 2-D image coordinate x is lexicographically ordered, the (x, y) th element of the Hessian matrix of $L(\hat{d}, \hat{f} | \hat{d})$ is given by its second-order derivative

$$\begin{aligned} \frac{\partial^2 L(\hat{d}, \hat{f} | \hat{d})}{\partial \hat{f}(x) \partial \hat{f}(y)} &= \sum_{z \in \Omega} w_1(z) \hat{d}(z-x) \hat{d}(z-y) \\ &+ \lambda \sum_{z \in \Omega} w_2(z) c(z-x) c(z-y), \quad x, y \in \Omega. \end{aligned} \quad (61)$$

If we set $w_1(z) = w_2(z) = 1, z \in \Omega$, then (61) becomes

$$\begin{aligned} \frac{\partial^2 L(\hat{d}, \hat{f} | \hat{d})}{\partial \hat{f}(x) \partial \hat{f}(y)} &= \hat{d}(x-y) * \hat{d}(-(x-y)) \\ &+ \lambda c(x-y) c(-(x-y)), \quad x, y \in \Omega. \end{aligned} \quad (62)$$

If periodic boundary condition is assumed, (62) indicates that the Hessian matrix is block circulant and its eigenvalues are (31), [26] the 2-D DFT of

$$\hat{d}(x) * \hat{d}(-x) + \lambda c(x) c(-x), \quad x \in \Omega. \quad (63)$$

Similarly, If the 2-D PSF coordinate x is lexicographically ordered, the (x, y) th element of the Hessian matrix of $L(\hat{d}, \hat{f} | \hat{f})$ is given by its second order derivative

$$\begin{aligned} \frac{\partial^2 L(\hat{d}, \hat{f} | \hat{f})}{\partial \hat{d}(x) \partial \hat{d}(y)} &= \sum_{z \in \Omega} w_1(z) \hat{f}(z-x) \hat{f}(z-y) \\ &+ \lambda \sum_{z \in \mathcal{D}} w_2(z) a(z-x) a(z-y), \quad x, y \in \mathcal{D}. \end{aligned} \quad (64)$$

If we set $w_1(z) = 1, z \in \Omega$ and $w_2(z) = 1, z \in \mathcal{D}$, then (64) becomes

$$\begin{aligned} \frac{\partial^2 L(\hat{d}, \hat{f} | \hat{f})}{\partial \hat{d}(x) \partial \hat{d}(y)} &= \hat{f}(x-y) * \hat{f}(-(x-y)) \\ &+ \lambda a(x-y) a(-(x-y)), \quad x, y \in \mathcal{D}. \end{aligned} \quad (65)$$

If periodic boundary condition is assumed, (65) indicates that the Hessian matrix is block circulant and its eigenvalues are (37), [26] the 2-D DFT of

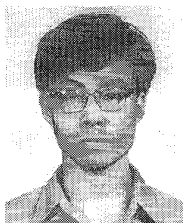
$$\hat{f}(x) * \hat{f}(-x) + \lambda a(x) a(-x), \quad x \in \mathcal{D}. \quad (66)$$

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Yu-Li You received the B.S. and M.S. degrees in electronic engineering from Xidian University (Northwestern Telecommunication Engineering Institute), Xi'an, China, in 1985 and 1988, respectively. Currently he is working toward the Ph.D. degree in electrical engineering at the University of Minnesota, Minneapolis, MN.

From 1988 to 1990 he was an electronics engineer at Guangzhou Institute of Electronic Technology, Chinese Academy of Sciences. He was a research assistant in the Department of Chemical Engineering, University of Virginia at Charlottesville, VA, from 1990 to 1991. His current research interests include image and video compression, image enhancement, image restoration, image analysis, communications, and neural networks.

Mr. You received the 1990 Science and Technology Award from Guangdong Provincial Government of China.



M. Kaveh (F'88) received the B.S. and Ph.D. degrees in electrical engineering from Purdue University, West Lafayette, IN, in 1969 and 1974, respectively, and the M.S. degree from the University of California at Berkeley, CA, in 1970.

He has been at the University of Minnesota since 1975, where he is presently a professor and head of the Department of Electrical Engineering. He is Vice President for Publications of the IEEE Signal Processing Society and was the General Chairman of ICASSP93.

Dr. Kaveh received the 1986 ASSP Senior (best paper) Award (with A. Barabell) and the 1988 ASSP Meritorious Service Award.